

Employee-treatment language across 133 technology acquisitions

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1. What this is

133 completed technology acquisitions whose SEC filings actually contain employee-related language, 13,817 passages drawn from those filings, and an unsupervised model run over that text to find the themes that recur across deals.

The question behind it: when companies buy other companies, what do they put in writing about the people they are acquiring? Not what they say publicly, and not what happened afterwards, but what the binding documents disclose.

Two things are worth saying at the start. Finding these deals was most of the work, because the public record is far thinner than the deal record, and the reasons are documented in section 2. And the model's output is a description of recurring contract language, not a finding about employees; section 11 is the boundary and it is not decoration.

2. How these deals were found

Most acquisitions leave no usable employee record in EDGAR. The buyer may be private, foreign, or a fund that never files, or the filing may exist without an employee-matters article. Selecting deals by what the target does finds companies; selecting them by what the buyer filed finds evidence. This sample is built the second way, and every deal that fell out is counted below rather than quietly dropped.

Step	Deals	What happened
Thomson/SDC deal catalog	26,369	Every transaction in the linked export
Acquirer not resolvable on EDGAR	-21,651	Private, private-equity, or foreign buyers that do not file
Target outside the technology screen	-3,658	Target SIC not in the 24-code digital-technology list
Candidate pool	1,060	SEC-registrant buyer with a technology target
Probed against EDGAR	1,060	Submissions index read for each buyer
filed a transaction agreement (EX-2)	89	The merger or purchase agreement itself
filed a merger proxy or tender offer	81	Employee and director interest sections
filed only an announcement	724	Press release in the window
filed nothing in the window	166	Dropped
Queued for retrieval	133	Ranked by disclosure richness
Retrieved and screened	400	Documents parsed, employee screen applied

Step	Deals	What happened
met the yield gate	133	≥ 10 passages from ≥ 2 documents
below the yield gate	102	Too little text to contribute
retrieved but no employee passage	165	Reported, not modelled
Deals in the model	133	Carrying 13,817 employee passages

Machine corroboration: for 112 deals the buyer's own filing names the target, which confirms both the transaction and the company match from the filer's document rather than from name similarity. That is a machine check, not human review.

3. The deals

133 transactions cleared the yield gate. The table lists the 40 largest by employee-passage count; the complete list is in `frozen_sample.csv`.

#	Acquirer	Target	Announced	Value (\$M)	Filing found	Passages
1	FiscalNote Holdings Inc	Aicel Technologies Inc	2021-12-29	not disclosed	proxy / tender offer	797
2	CF Acquisition Corp VI	Rumble Inc	2021-12-02	3,150.00	proxy / tender offer	708
3	ADTRAN Inc	ADVA Optical Networking SE	2021-08-30	1,026.046	agreement (EX-2)	534
4	System1 LLC	Protected.Net Grp Ltd	2021-06-29	not disclosed	proxy / tender offer	492
5	Ginkgo Bioworks Holdings Inc	Baktus Inc-Epidemiological	2022-08-19	not disclosed	proxy / tender offer	476
6	Take-Two Interactive Software	Zynga Inc	2022-01-10	12,382.165	agreement (EX-2)	418
7	Advanced Micro Devices Inc	Xilinx Inc	2020-10-27	35,728.81	proxy / tender offer	403
8	Entegris Inc	CMC Materials Inc	2021-12-15	5,729.282	proxy / tender offer	314
9	Advent Technologies Holdings	Ultracell LLC	2021-02-18	not disclosed	proxy / tender offer	308
10	Teledyne Technologies Inc	FLIR Systems Inc	2021-01-04	7,500.665	proxy / tender offer	287
11	Unity Software Inc	Ironsource Ltd	2022-07-13	4,422.114	proxy / tender offer	277

#	Acquirer	Target	Announced	Value (\$M)	Filing found	Passages
12	S&P Global Inc	IHS Markit Ltd	2020-11-30	43,478.099	proxy / tender offer	259
13	Analog Devices Inc	Maxim Integrated Products Inc	2020-07-13	21,290.859	proxy / tender offer	227
14	Teladoc Health Inc	Livongo Health Inc	2020-08-05	17,539.058	proxy / tender offer	218
15	EVgo Inc	Recargo Inc	2021-07-09	25.00	proxy / tender offer	217
16	Uber Technologies Inc	Postmates Inc	2020-07-06	2,650.00	proxy / tender offer	214
17	Rocket Lab USA Inc	Advanced Solutions Inc	2021-10-12	45.50	proxy / tender offer	213
18	ViaSat Inc	Euro Broadband Infrastructure	2020-11-19	not disclosed	proxy / tender offer	208
19	Skillsoft Corp	Ryzac Inc	2021-12-22	525.00	proxy / tender offer	193
20	Intuit Inc	Credit Karma Inc	2020-02-24	7,100.00	proxy / tender offer	187
21	Opendoor Technologies Inc	Pro.com Home Services LLC	2021-09-07	not disclosed	proxy / tender offer	159
22	Bally's Corp	Bet.Works Corp	2020-11-18	139.382	agreement (EX-2)	147
23	Intercontinental Exchange Inc	Engaged Trckng (Et) Index Ltd	2022-07-21	not disclosed	proxy / tender offer	135
24	Rekor System Inc	Waycare Technologies Ltd	2021-08-19	60.121	announcement	131
25	Lumentum Holdings Inc	NeoPhotonic s Corp	2021-11-04	863.571	agreement (EX-2)	130
26	Emerson Electric Co	Aspen Technology Inc	2021-10-11	10,864.939	agreement (EX-2)	121
27	Rocket Lab USA Inc	SolAero Technologies Corp	2021-12-13	80.00	proxy / tender offer	121
28	Hewlett Packard Enterprise Co	Silver Peak Systems Inc	2020-07-13	925.00	agreement (EX-2)	118
29	Marvell Technology Inc	Innovium Inc	2021-08-03	1,153.859	proxy / tender offer	113
30	8x8 Inc	Fuze Inc	2021-12-01	250.00	agreement (EX-2)	113

#	Acquirer	Target	Announced	Value (\$M)	Filing found	Passages
31	Clarivate PLC	ProQuest LLC	2021-05-17	6,336.023	agreement (EX-2)	111
32	Goldman Sachs Group Inc	Greensky Inc	2021-09-15	2,239.124	proxy / tender offer	107
33	MKS Instruments Inc	Photon Control Inc	2021-05-10	311.903	announcement	102
34	Omnicell Inc	Pharm Strategies Group-Business	2020-08-12	225.00	agreement (EX-2)	100
35	Flotek Industries Inc	JP3 Measurement LLC	2020-05-18	42.80	agreement (EX-2)	95
36	Ouster Inc	Sense Photonics Inc	2021-10-05	68.02	proxy / tender offer	94
37	DocuSign Inc	Seal Software Ltd	2020-02-27	188.00	agreement (EX-2)	93
38	Advent Technologies Holdings	Fischer Eco Solutions GmbH	2021-06-25	not disclosed	proxy / tender offer	90
39	Stryker Corp	Vocera Communications Inc	2022-01-06	2,970.467	agreement (EX-2)	88
40	Moody's Corp	Risk Management Solutions Inc	2021-08-05	1,978.613	agreement (EX-2)	86

A further 165 deals were retrieved and produced no employee passage at all. They stay in the record. A filed agreement without employee language is a fact about disclosure practice, not evidence that the transaction had no employee arrangements.

4. The corpus the model reads

A passage is one block of text from a transaction-linked SEC document that mentions employees, compensation, benefits, equity, retention, severance, or employment terms. Exact duplicates collapse to one row and near-identical legal boilerplate is grouped into provision families, so one heavily repeated clause cannot dominate the model.

Measure	Count
Documents parsed	6,156
Transaction-linked documents kept	2,862
Documents excluded as unrelated to the deal	3,294
Candidate passages screened	42,235
Passages included	16,079
Passages excluded	26,156

Measure	Count
Provision families	34,295
Largest single deal's share of modelled passages	5.8%

The screen rejects far more than it keeps, and deliberately so: navigation fragments, accounting context, safe-harbour boilerplate, and bare captions all mention employees without saying anything about how they are treated. Every rejection reason is counted in `corpus_manifest.json`.

Where the text comes from:

Filing type	Passages	Share
EX-2.1	4,821	30.0%
424B3	3,475	21.6%
EX-10.1	1,944	12.1%
S-4/A	1,360	8.5%
S-4	1,286	8.0%
EX-99.1	678	4.2%
DEFM14A	471	2.9%
8-K	360	2.2%

5. What the unsupervised model found

The model is given no categories. It reads word and two-word patterns across the passages, factorises them into a fixed number of components, and gives every passage a weight on each. The names below were written after reading the top terms and the highest-weighted passages of each component. They are descriptions of what the model grouped, not labels it was taught, and they remain provisional until two reviewers score them independently.

The model is fitted on the 133 deals of the frozen sample. Its assignments are then projected onto every passage in the corpus, including deals that fell below the yield gate, which is why a component's passage count spans more deals than the sample itself. The deal counts below are for the frozen sample.

Executive and officer language

Defining words: executive, officer, chief, employment, executive officer, chief executive, directors, employees, board, officers

Reading (provisional): References to executives, officers, directors and the board. This component mixes two things rather than one: contractual executive-employment and change-in-control terms, and the officer or director language that appears in proxies and announcement press releases, including quoted remarks by a chief executive. Its internal coherence is the lowest of the three, which is what a mixed component looks like. It should not be read as a single kind of employee provision.

Measure	Value
Passages where this is the strongest theme	5,741
Distinct provision families	4,947
Deals in the sample carrying it	133 of 133

Measure	Value
Internal coherence	0.258
Leave-one-deal-out recovery	0.864

Stability reading: this component reproduces when any single deal is removed.

"We had a strong start to the year as we continue to accelerate innovation on our A.I.-driven expert platform," said Sasan Goodarzi, Intuit's chief executive officer. "We delivered double-digit revenue growth in the quarter and are excited by the velocity of our innovation." (source)

- Visionary, Founder-Led Management Team. Our co-founder and CEO, Henry Schuck, pioneered the category of go-to-market intelligence and is the driving force behind our vision, mission, and culture. Our highly talented, customer-centric senior leadership enables us to rapidly develop new products, move more quickly than our competition, and build our fast-paced, execution-oriented culture. (source)

"We are delighted at the prospect of welcoming the Redflex team and their customers to the Verra Mobility family," said David Roberts, Chief Executive Officer, Verra Mobility. "We are incredibly excited for this step forward in building the Verra Mobility of the future, expanding our portfolio of safe city solutions, and solidifying our position as a global leader in smart transportation." (source)

Benefit plans and ERISA

Defining words: plan, benefit, erisa, employee, benefit plan, code, pension, employee benefit, plans, benefit plans

Reading (provisional): Definitions and covenants about employee benefit plans: plan schedules, ERISA representations, pension and welfare arrangements, and undertakings about benefits after closing. This is the most self-consistent contractual component.

Measure	Value
Passages where this is the strongest theme	5,762
Distinct provision families	5,016
Deals in the sample carrying it	129 of 133
Internal coherence	0.351
Leave-one-deal-out recovery	0.889

Stability reading: this component reproduces when any single deal is removed.

"Benefit Plans" means the Company Benefit Plans and the ADP Benefit Plans. (source)

(v) As of the Closing Date, there are no Canadian Pension Plans. (source)

"Plan" means any employee benefit plan as defined in Section 3(3) of ERISA, including any employee welfare benefit plan (as defined in Section 3(1) of ERISA), any employee pension benefit plan (as defined in Section 3(2) of ERISA), and any plan which is both an employee welfare benefit plan and an employee pension benefit plan, and in respect of which the Borrower or any ERISA Affiliate is an "employer" as defined... (source)

Equity awards at the effective time

Defining words: stock, shares, common, common stock, restricted, restricted stock, units, options, merger, outstanding

Reading (provisional): Treatment of stock, options, restricted stock and restricted stock units when the transaction closes: conversion, assumption, vesting and cash-out of outstanding awards. It is the most stable

component, recovering in nearly every leave-one-deal-out fold.

Measure	Value
Passages where this is the strongest theme	4,576
Distinct provision families	3,725
Deals in the sample carrying it	118 of 133
Internal coherence	0.432
Leave-one-deal-out recovery	0.983

Stability reading: this component reproduces when any single deal is removed.

Assumed Company RSU (source)

Assumed RSU (source)

Converted Parent RSU (source)

6. The AI subgroup

The earlier version of this study screened for AI first and then looked for filings. That produced thirteen usable deals, because the companies an AI keyword finds are mostly bought by firms that never file with the SEC. Here the label is applied afterwards, to deals already known to have employee disclosure, which turns it into a question that can be answered: among transactions whose employee terms are public, how many describe the target in AI terms?

Label	Deals
Filings describe the target in explicit AI terms	17
Weaker or adjacent AI language	0
No AI language near the target's name	116
Total in the sample	133

Team-joining language appears in 45 deals and explicit acqui-hire language in 3.

The AI-labelled deals, with the wording their own filings use:

Acquirer	Target	Wording found
ADTRAN Inc	ADVA Optical Networking SE	machine learning
CF Acquisition Corp VI	Rumble Inc	artificial intelligence
Certara Inc	Pinnacle 21 LLC	machine learning
DLH Holdings Corp	Irving Burton Associates LLC	artificial intelligence
DocuSign Inc	Seal Software Ltd	ai, artificial intelligence, machine learning, natural lang...
Hewlett Packard Enterprise Co	Silver Peak Systems Inc	ai, self-driving
Intercontinental Exchange Inc	Ellie Mae Inc	artificial intelligence, machine learning
Mitek Systems Inc	HooYu Ltd	artificial intelligence, computer vision
Opendoor Technologies Inc	Pro.com Home Services LLC	machine learning

Acquirer	Target	Wording found
Progress Software Corp	Chef Software Inc	machine learning
Quantum Computing Inc	QPhoton Inc	computer vision
Quantum Corp	Sq Box Sys Ltd	artificial intelligence, machine learning
Rekor System Inc	Waycare Technologies Ltd	ai, artificial intelligence
SentinelOne Inc	Attivo Networks Inc	ai-powered
Smith Micro Software Inc	Avast Plc-Family Safety	artificial intelligence, machine learning
Take-Two Interactive Software	Zynga Inc	machine learning
ViaSat Inc	Euro Broadband Infrastructure	artificial intelligence, machine learning

Every label here is machine-derived and pending human review. A deal marked with no AI language is a deal whose retrieved filings do not describe it that way; it is not a finding that the target does no AI work.

7. Does the result depend on how we built it?

The bounded fit sample can be spread evenly across deals, across document families, or not balanced at all. The primary setting was fixed before this run. Re-fitting under the other two is the check that the components are a property of the text rather than of that choice.

Fit balance	Components	Recovery per component	Leading terms
source_family (primary)	3	0.86, 0.89, 0.98	executive; plan; stock
deal	3	0.95, 0.95, 0.95	employment; stock; plan
none	3	0.90, 0.94, 0.98	executive; plan; stock

All three settings return the same three themes in different proportions, and every component recovers well above the 0.80 floor. The themes are not an artefact of the balancing choice.

8. Tone, as a secondary diagnostic only

This counts protective and negative wording per hundred tokens and subtracts the average for the same filing type, so deals are compared against ordinary legal language rather than against plain English. It measures how documents are written. It is not evidence that any buyer treated people better.

Interpretation status recorded in the manifest: secondary_diagnostic_corpus_not_validated.

Deal	Protective-language residual
Opendoor Technologies Inc / Commeasure Pte Ltd	+5.330
Vontier Corp / DRB Systems LLC	+1.826
KKR & Co Inc / Therapy Brands Holdings LLC	+1.570
HealthStream Inc / myClinicalExchange LLC	+1.374
Mastech Digital Inc / AmberLeaf Partners Inc	+1.060
...	

Deal	Protective-language residual
CleanSpark Inc / P2klabs Inc	-0.877
Workiva Inc / Parseport ApS	-0.888
Simulations Plus Inc / Lixoft SAS	-0.893

Only deals in the frozen sample are named. A high residual means the filing uses more protective wording than is typical for that filing type, and nothing more.

9. Which checks the model passed and failed

The pipeline runs its own checks and reports them whatever they say. Reporting only the checks that passed would make the weak parts of this result invisible.

Automated checks: 71 passed, 0 failed, 1 warning.

Warning	Value	What it means
agglomerative adjusted rand	0.03419036008	Identical fit universe; L2 word/bigram TF-IDF; sklearn cosine metric and average linkage; ARI is permutation-invariant, so label alignment is not req...

The agglomerative comparison deserves plain words. A second, unrelated clustering method was run over the same passages and its groups were compared with the model's. The agreement is low. The three components are individually stable, reproducing when any single deal is dropped, but a different algorithm would not carve the text the same way. Read the components as recurring language, not as the only true division of it.

Model status recorded in the manifest: modeled. That word means the model ran and its own checks are recorded. It does not mean the corpus was validated by a person; section 11 says what was not done.

10. Reproduction

Every table above is generated from committed code and frozen artifacts:

```
tag-edgar screen-disclosure-pool data/derived/deal_catalog.csv
tag-edgar probe-disclosure data/derived/disclosure_pool/pool.csv
tag-edgar build-disclosure-queue data/derived/disclosure_probe/probe_results.csv
tag-edgar run-disclosure-sample data/derived/disclosure_review_queue.csv
tag-edgar build-employee-corpus <queue> data/derived/disclosure_runs
tag-edgar freeze-disclosure-sample <queue> <passages> data/derived/disclosure_runs
tag-edgar analyze-employee-topics <queue> <corpus>
python scripts/build_disclosure_sample_report.py
```

Selection rule disclosure-first-v1; corpus hash 97ea426f6900a345...

The tables themselves are published under data/published/disclosure_sample_133/, so any number here can be checked against the file it came from without rerunning the pipeline. The passage corpus, the retrieved documents, and the SDC archive are not published; that directory's README gives the reason for each.

11. What this cannot show

- **The sample is selected by disclosure.** It describes acquisitions whose buyers

file with the SEC and put the agreement on the record. Deals by private, private-equity, and foreign buyers are largely absent. That is a property of the public record, not a sampling choice that can be corrected by weighting.

- **These are contracts and proxies, not outcomes.** A clause promising benefit

continuity is a promise. Nothing here shows whether any employee stayed, was paid what was promised, or was satisfied.

- **Nothing here is causal.** The clusters describe how transaction documents are

written. They cannot say that a drafting choice caused an employee result.

- **The cluster names are provisional.** They were written after seeing the model's

output and have not been scored by two independent reviewers.

- **Corpus relevance audit: not run for this cycle.** The screen that decides which passages

count as employee-related has not been validated by a human on this corpus, so the proportion of included passages that a person would call relevant is unmeasured. The earlier cycle-4 audit of a different corpus scored 72% against a 90% threshold, so this is a real and quantified risk, not a formality.

- **Company matching is machine-confirmed.** The buyer's filing naming the target is

strong corroboration, but no person has checked each pairing.